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Knowledge Representation

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

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Representation

AI agents deal with knowledge (data)

- ▶ Facts (believe & observe knowledge)
- ▶ Procedures (how to knowledge)
- ▶ Meaning (relate & define knowledge)

Right representation is crucial

- ▶ Early realization in AI
- ▶ Wrong choice can lead to project failure
- ▶ Active research area

Some General Representations

- ▶ Expert Systems
- ▶ Logical Representations
- ▶ Production Rules
- ▶ Semantic Networks
 - ▶ Conceptual graphs, frames
- ▶ Description Logics

What is an Expert System?

A rule based computer system which simulates a human expert in his/her field of expertise in an attempt to solve a particular problem.

An expert system (commonly referred to as a knowledge based system):

- ▶ Accumulates the knowledge of human experts or the knowledge obtained from a variety of different “expert” sources (e.g. documentation).
- ▶ The decisions (conclusions) of expert systems are based on:
 - ▶ The accumulated knowledge (represented as facts)
 - ▶ Rules operating on the existing

Intelligence Vs Expertise

Expertise and intelligence are not the same things (although they are related).

Expertise requires long time to learn (e.g. it takes 6 years to become a doctor).

Expertise is a large amount of knowledge (in some domain).

Expertise is easily recalled.

Intelligence allows you to use your expertise (apply the knowledge).

Expertise enables you to find solutions much faster.

Characteristics of an Expert System

Goal oriented, i.e. their actions (questions to the user) are based on the desired goal(s) that the system must achieve.

Able to work with imperfect information. If some specific information is missing the expert system must choose an alternative route (question to the user) to reach a decision.

Able to cope with different combinations of events, independent of the order that these appear.

Able to justify (give the reasons) for their decision/recommendation.

Characteristics of an Expert System

To be able to replace an expert with a computerized expert system, the expert must:

- ▶ be able to perform the task;
- ▶ know how they perform the task;
- ▶ be able to explain how they perform the task;
- ▶ have the time to describe in detail how they perform the task;
- ▶ be willing to explain how they perform the task.

The above are included in the limitations (disadvantages) of expert systems.

An example of facts and rules

Facts:

- ▶ Engine off.
- ▶ Pedestrian crossing the road in a distance less than 10 meters away.

Rules:

- ▶ if (car needs to be driven) and (engine off)
then start engine
- ▶ if (a pedestrian is crossing the road) and
(in a distance less than 10 meters away)
then break

Example – Diagnosis of Car Problems over the Phone

Mechanic: Hello, how may I help?

Driver: My car wouldn't start.

Mechanic: What happens when you turn the key in the ignition to try to start the car?

Driver: It turns over OK, but it just won't start.

Mechanic: Have you checked that you aren't out of petrol?

Driver: I'm not certain the tank is empty.

Example – Diagnosis of Car Problems over the Phone

Mechanic: As you crank the starter, do you smell petrol?

Driver: No, I turned it over for a long time, but didn't smell anything

Mechanic: Based on what you've told me, I'm almost certain your car is out of petrol.

Driver: Thanks for the advice. Mind telling me how you reached your conclusion?

Mechanic: When a car won't start my initial suspicion is that the battery is dead, the starter has failed or some other electrical problem exists. Your input that the starter operates makes it more likely that no fuel is getting to the engine. Although you are not sure that the petrol tank is empty, the fact that you don't smell petrol when the engine turns over supports my conclusion that you are out of petrol.



Early Expert Systems

Production systems (or rule-based systems) are programs that instead of conventional algorithms use sets of IF-THEN rules (production rules). Unlike in algorithms, the order in which these rules should be used is not specified. It is decided by the program itself with respect to a problem state.

In 1943, Post proved that any computable problem can be implemented in a production system

Early Expert Systems

In 1961, A. Newell and H. Simon wrote a program called General Problem Solver (GPS) that could solve many different problems using only a small set of rules.

- ▶ GPS used a strategy known as means–ends analysis.
- ▶ GPS produced solutions very similar to those people came up with.
- ▶ Methods that can be applied to a broad range of problems are called weak methods (because they use weak information about the problem domain). Their performance, however, is also usually weak.

Knowledge-Based Systems

DENDRAL (Feigenbaum et al, 1969) was a program that used rules to infer molecular structure from spectral information. The challenge was that the number of possible molecules was so large, that it was impossible to check all of them using simple rules (weak method).

- ▶ The researchers consulted experts in chemistry and added several more specific rules to their program. The number of combinations the program had to test was reduced dramatically.
- ▶ DENDRAL demonstrated the importance of the domain-specific knowledge.

Knowledge-Based Systems

MYCIN developed in the 1970s was an expert system which is responsible for the diagnosis and recommendation of treatment for some blood infections.

- ▶ Its knowledge was represented as a set of rules, similar to:
 - ▶ if the infection is x and the morphology of the organism is y and the aerobicity of the organism is z then there is suggestive evidence (0.8) that infection is of w type and that the class of the organism is v .
 - ▶ 0.8 describes the probability that the conclusion is certain.
- ▶ MYCIN was written in Lisp, but it was never placed in practice for ethical and legal reasons

Advantages/Limitations of Expert Systems

Advantages of Expert Systems over Human Experts

- ▶ An expert system can operate constantly 24 hours per day.
- ▶ An expert system can exceed the performance of any human expert, as it can combine knowledge from a number of different experts.

Some of the limitations of expert systems include:

- ▶ Not able to cope with unseen information.
- ▶ Not able to learn independently in a similar manner that humans learn. They need to be programmed in advance.
- ▶ Not able to cope with noise.
- ▶ Not able to adapt to new environments.

What is a Logic?

A language with concrete rules

- ▶ No ambiguity in representation (may be other errors!)
- ▶ Allows unambiguous communication and processing
- ▶ Very unlike natural languages e.g. English

Many ways to translate between languages

- ▶ A statement can be represented in different logics
- ▶ And perhaps differently in same logic

Expressiveness of a logic

- ▶ How much can we say in this language?

Not to be confused with logical reasoning

- ▶ Logics are **languages**, reasoning is a **process** (may use logic)

Syntax and Semantics

Syntax

- ▶ Rules for constructing legal sentences in the logic
- ▶ Which symbols we can use (English: letters, punctuation)
- ▶ How we are allowed to combine symbols

Semantics

- ▶ How we interpret (read) sentences in the logic
- ▶ Assigns a meaning to each sentence

Example: “All lecturers are two meter tall”

- ▶ A valid sentence (syntax)
- ▶ And we can understand the meaning (semantics)
- ▶ This sentence happens to be false (there is a counterexample)

Propositional Logic

Syntax

- ▶ Propositions, e.g. “it is wet”
- ▶ Connectives: and, or, not, implies, iff (equivalent)

$\wedge \vee \neg \rightarrow \leftrightarrow$

- ▶ Brackets, $\top$ (true) and $\perp$ (false)

Semantics (Classical AKA Boolean)

- ▶ Define how connectives affect truth
 - ▶ “P and Q” is true if and only if P is true and Q is true
- ▶ Use truth tables to work out the truth of statements

Predicate Logic

Propositional logic combines atoms

- ▶ An atom contains no propositional connectives
- ▶ Have no structure (`today_is_wet`, `john_likes_apples`)

Predicates allow us to talk about objects

- ▶ Properties: `is_wet(today)`
- ▶ Relations: `likes(john, apples)`
- ▶ True or false

In predicate logic each atom is a predicate

- ▶ e.g., first order logic, higher-order logic

First Order Logic

More expressive logic than propositional

Constants are objects: john, apples

Predicates are properties and relations:

- ▶ likes(john, apples)

Functions transform objects:

- ▶ likes(john, fruit_of(apple_tree))

Variables represent any object: likes(X, apples)

Quantifiers qualify values of variables

- ▶ True for all objects (Universal):

$\forall X. \text{likes}(X, \text{apples})$

- ▶ Exists at least one object (Existential):

$\exists X. \text{likes}(X, \text{apples})$

Examples: FOL Sentences

“Every rose has a thorn”

$$\forall X.(\text{rose}(x) \rightarrow \exists Y.(\text{has}(X, Y) \wedge \text{thorn}(Y)))$$

For all X

- ▶ if (X is a rose)
- ▶ then there exists Y
 - ▶ (X has Y) and (Y is a thorn)

Examples: FOL Sentences

“On Mondays and Wednesdays I go to John’s house for dinner”

$$\forall X.((\text{is_mon}(X) \vee \text{is_wed}(X))) \rightarrow \text{eat_meal}(\text{me}, \text{house_of}(\text{jhon}), X)$$

Note the change from “and” to “or”

- Translating is problematic

Higher Order Logic

More expressive than first order

Functions and predicates are also objects

- ▶ Described by predicates: `binary(addition)`
- ▶ Transformed by functions: `differentiate(square)`
- ▶ Can quantify over both

E.g., define red functions as having zero at 17

$$\forall F.(\text{red}(F) \leftrightarrow F(0) = 17)$$

Much harder to reason with

Beyond True and False

Multi-valued logics

- ▶ More than two truth values
- ▶ e.g., true, false & unknown
- ▶ Fuzzy logic uses probabilities, truth value in $[0,1]$
 - ▶ Cox's Theorem

Modal logics

- ▶ Modal operators define mode for propositions
- ▶ Epistemic logics (belief)
 - ▶ e.g. necessarily p , possibly p , ...
- ▶ Temporal logics (time)
 - ▶ e.g. always p , eventually p , ...

Logic is a Good Representation

Fairly easy to do the translation when possible

Branches of mathematics devoted to it

It enables us to do logical reasoning

- ▶ Tools and techniques come for free

Basis for programming languages

- ▶ Prolog uses logic programs (a subset of FOL)
- ▶ λ Prolog based on HOL

Non-Logical Representations?

Production rules

Semantic networks

- ▶ Conceptual graphs
- ▶ Frames

Logic representations have restrictions and can be hard to work with

- ▶ Many AI researchers searched for better representations

Production Rules

Rule set of $\langle \text{condition}, \text{action} \rangle$ pairs

- ▶ “if condition then action”

Match-resolve-act cycle

- ▶ Match: Agent checks if each rule's condition holds
- ▶ Resolve:
 - ▶ Multiple production rules may fire at once (conflict set)
 - ▶ Agent must choose rule from set (conflict resolution)
- ▶ Act: If so, rule “fires” and the action is carried out

Working memory:

- ▶ rule can write knowledge to working memory
- ▶ knowledge may match and fire other rules

Production Rules Example

IF (at bus stop AND bus arrives) THEN action(get on the bus)

IF (on bus AND not paid AND have oyster card) THEN action(pay with oyster) AND add(paid)

IF (on bus AND paid AND empty seat) THEN sit down

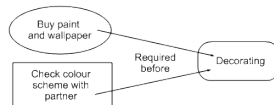
Conditions and actions must be clearly defined

- can easily be expressed in first order logic!

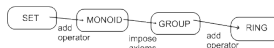
Graphical Representation

Humans draw diagrams all the time, e.g.

- Causal relationships

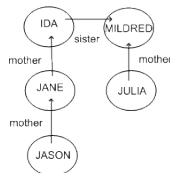
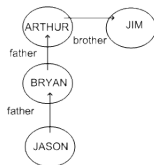


And relationships between ideas



Graphs easy to store in a computer

To be of any use must impose a formalism



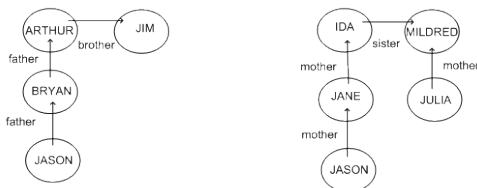
- Jason is 15, Bryan is 40, Arthur is 70, Jim is 74
- How old is Julia?

Semantic Networks

Because the syntax is the same

- We can guess that Julia's age is similar to Bryan's

Formalism imposes restricted syntax



Graphical representation (a graph)

- Links indicate subset, member, relation, ...

Equivalent to logical statements (usually FOL)

- Easier to understand than FOL?
- Specialised SN reasoning algorithms can be faster

Conceptual Graphs

Example: natural language understanding

- ▶ Sentences with same meaning have same graphs
- ▶ e.g., Conceptual Dependency Theory (Schank)

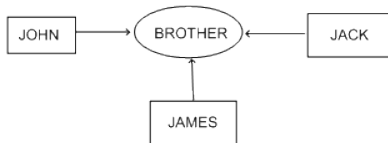
Semantic network where each graph represents a single proposition

Concept nodes can be

- ▶ Concrete (visualisable) such as restaurant, my dog Spot
- ▶ Abstract (not easily visualisable) such as anger

Edges do not have labels

- ▶ Instead, conceptual relation nodes
- ▶ Easy to represent relations between multiple objects



Frame Representations

Semantic networks where nodes have structure

- ▶ Frame with a number of slots (age, height, ...)
- ▶ Each slot stores specific item of information

When agent faces a new situation

- ▶ Slots can be filled in (value may be another frame)
- ▶ Filling in may trigger actions
- ▶ May trigger retrieval of other frames

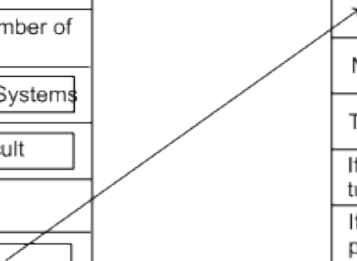
Inheritance of properties between frames

- ▶ Very similar to objects in OOP

Example: Frame Representation

Lecture	
Specialisation of: meeting	
Context: large number of students	
Course:	Op. Systems
Level:	Difficult
If difficult, then pay attention	
Lecturer:	
Room*:	

Lecturer	
Name:	Prof Jones
Tolerance:	Intolerant
If intolerant, then turn off mobile phone	
If intolerant, then pay attention	



Example: Frame Representation

Slots in a frame can contain

- ▶ Information for choosing a frame in a situation
- ▶ Relationships between this and other frames
- ▶ Procedures to carry out after various slots filled
- ▶ Default information to use where input is missing
- ▶ Blank slots: left blank unless required for a task
- ▶ Other frames, which gives a hierarchy

Can also be expressed in first order logic

Representation & Logic

AI wanted “non-logical representations”

- ▶ Production rules
- ▶ Semantic networks
 - ▶ Conceptual graphs, frames

But all can be expressed in first order logic!

Best of both worlds

- ▶ Logical reading ensures representation well-defined
- ▶ Representations specialised for applications
- ▶ Can make reasoning easier, more intuitive